

PROJECT PROPOSAL

PROJECT TITLE

EloChess: A Transformer-Based approach to predict the level of chess game play

DATA SET

<https://database.lichess.org> (Open-source dataset of ~100 million games played on lichess every month)

Our project will leverage the extensive Lichess database to extract meaningful features and sample games based on Elo ratings. The dataset is structured in PGN (Portable Game Notation) format, containing metadata such as player names, ratings, game results, openings, and timestamps, followed by a sequence of moves annotated with Stockfish evaluations. Key fields include *WhiteElo* and *BlackElo*, which represent the skill levels of the players, as well as *Opening*, *Termination*, and *TimeControl*, which provide contextual insights into different playing styles. To capture the strength and accuracy of moves, we will extract centipawn loss (CPL) values using Stockfish, measuring how much each move deviates from optimal play. This will serve as a crucial feature in evaluating a player's consistency and tactical sharpness. By sampling games across different rating brackets, we will ensure a diverse and balanced dataset, allowing our transformer model to learn stylistic and strategic variations across varying skill levels.

PROJECT IDEA

Our project aims to predict the skill level of chess players based on their in-game moves using transformer-based models. By analysing sequences of moves in chess games, we seek to develop a model that can accurately predict the level of play in terms of the **Elo** rating. We intend to leverage the power of transformers to process entire game sequences as a series of tokens, capturing deep positional and strategic patterns combined with stockfish analysis of individual games. This approach enables the model to learn nuanced relationships between moves, uncovering stylistic and tactical differences that distinguish players of varying expertise.

To achieve this, we will preprocess chess game data by encoding move sequences in a format compatible with transformers, such as PGN (Portable Game Notation) converted into tokenized inputs. We will fine-tune a transformer model, potentially using architectures like BERT or GPT adapted for sequential chess data, to predict a player's Elo rating. The model will be trained on large-scale datasets containing games from online chess platforms, ensuring exposure to diverse playing styles. By leveraging attention mechanisms, our approach can identify key moments in a game that signal a player's proficiency. The goal is to create a robust and interpretable model that can assess player strength dynamically, with potential applications in chess coaching, matchmaking, and automated analysis.

SOFTWARE

Game Analysis	Prediction/Modellig	Dev Environment
○ Stockfish ○ AlphaZero	○ Transformers ○ LSTM	○ Linux/macOS ○ VSCode ○ Cuda

DETAILS OF COMPUTING RESOURCES

Need: NVIDIA GeForce MX230 (used for LSTM based approach)

Have: NVIDIA GeForce RTX4060 Laptop (Can be enough for tiny BERT)

RELEVANT RESEARCH PAPERS

1. Omori, M., & Tadeipalli, P. (2024). *Chess rating estimation from moves and clock times using a CNN-LSTM*. arXiv preprint arXiv:2409.11506.
2. T. Tijhuis, P. M. Blom and P. Spronck, "Predicting Chess Player Rating Based on a Single Game," 2023 IEEE Conference on Games (CoG), Boston, MA, USA, 2023, pp. 1-8, doi: 10.1109/CoG57401.2023.10333133.
3. Acher, M. (2016). Large-scale analysis of chess games with chess engines: A preliminary report. *arXiv:1607.04186*. <https://arxiv.org/abs/1607.04186>.

TEAM

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MILESTONE

Halfway Milestone Report

By the halfway point of our project, we aim to have completed the key foundational steps required for training our transformer model. This includes thorough data analysis, preprocessing, feature extraction, and tokenization of chess game sequences. We will have explored the Lichess database, extracted relevant game data, and structured it according to Elo rating brackets to ensure a balanced representation of skill levels. Our preprocessing pipeline will involve cleaning and standardizing game records, handling missing or inconsistent data, and converting PGN (Portable Game Notation) into a format suitable for transformer-based models. Additionally, we will have implemented an effective tokenization strategy to represent chess moves as model-readable sequences, preserving the sequential and contextual aspects of gameplay. This stage will also involve exploratory data analysis (EDA) to understand move distributions, player styles, and rating trends, providing insights into feature importance. By this milestone, our dataset should be well-structured and ready to be fed into a transformer model for training and evaluation in the subsequent phases.